

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

steering_knuckle_RH

Commodity: CAST AND MACHINE · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 6 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥198.06	16.6%	37.8%	16.6%	7.4%

Model Confidence: **Low** · 6 traced operations · 15 data points auditable**Engine Warnings (1)**

rawMaterial.materialId: Material rate confidence: Low

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 356.1 cm³; mass for the selected material family 2.564 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: cast and machine

Alloy / material: EN-GJL-250 · Net weight: 2.625 kg (measured)

Material utilisation: 65% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 6

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥32.81	16.6%	
2. Process (Machine)	¥74.95	37.8%	
3. Direct Labour	¥32.97	16.6%	
4. Tooling (amortised)	¥19.55	9.9%	
5. Packaging	¥1.45	0.7%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
6. Logistics	¥2.44	1.2%	
Factory Cost	¥164.16	82.9%	
7. Overhead (SG&A)	¥19.23	9.7%	
Subtotal	¥183.39	92.6%	
8. Supplier Margin	¥14.67	7.4%	
TOTAL SHOULD-COST	¥198.06	100.0%	

§2 — Commercial Parameters

Overhead Rate	12.0%	Supplier Margin	8.0%
Packaging / part	¥1.45	Logistics / part	¥2.44
Total Tooling Cost	¥3909502.26	Amortisation Volume	200,000 parts

Geometric DFM / DFA - measured from the CAD model

7 issue(s) across 117 instance(s), from 172 measured feature(s) and 7 rule(s). Every issue names the B-rep faces that produced it and the published source of its threshold. Priced findings total ¥57.77/part; issues shown without a figure are quality or yield risks with no modelled cost path — see each entry.

[MINOR] Internal corner radius below any economic end mill (14 instances)

¥47.46/part

Worst: Internal corner at face 100 is R0.25 mm.

measured radiusMm 0.25-0.25mm threshold < 1mm

Faces: 89, 94, 96, 98, 100, 102, 106, 108, 110, 112, 114, 116, 118, 120

Fix: Open the corner to at least R1 mm so a standard end mill can cut it; below that expect a slow small-diameter cutter, or EDM.

Source: Machinery's Handbook — end milling: cutter diameter and corner radii

An internal corner cannot be smaller than the cutter that makes it. Below R1 mm the cutter is fragile and slow; the corner is then usually EDM or a broach.

[ADVISORY] Non-preferred hole diameter — special tool (4 instances)

¥6.81/part

Worst: # 46.00 mm is not a standard drill size; nearest is # 45 mm.

measured diaMm 0.3-46mm threshold > 45mm

Faces: 73, 75, 77, 79, 81, 83, 85, 87, 92, 104, 288

Fix: If the fit allows, move to # 45 mm and save a special tool or a separate bore/ream cycle.

Source: Machinery's Handbook — standard twist-drill diameters (metric series)

A diameter off the standard series needs a special or a bore/ream cycle instead of a single drilled pass. Tolerance ±0.05 mm on the match.

[MINOR] Hole deeper than standard drill reach

¥3.49/part

Worst: # 4.0 mm hole is 24.9 mm deep — 6.2:1 depth-to-diameter.

measured IdRatio 6.237:1 threshold > 5:1

Faces: 143, 167

Fix: Beyond ~5:1 needs peck-drilling or extended-reach tooling — slower cycle, higher tool cost and greater breakage risk. Open the diameter if the function allows.

Source: Machinery's Handbook — drilling: depth-to-diameter limits and standard drill sizes

Standard jobber drills reach ~5×D; parabolic / coolant-through tooling reaches ~10×D; beyond that gun drilling. Reported at the point the process and the cost step.

[MAJOR] Wall below minimum draft for the casting route (60 instances)

Worst: Face 1 draws at 0.00° against a 0.5° minimum for this route.

measured draftDeg 0-0° threshold < 0.5°

Faces: 1, 4, 9, 12, 18, 19, 22, 23, 31, 32, 33, 34, 46, 47, 48, 49, 53, 57, 60, 71, 88, 90, 97, 105, 113, 123, 124, 131, 132, 134 ... (60 total)

Fix: Add draft to at least 0.5° per side, or accept the ejection drag and the die wear it causes.

Source: ASM Handbook Vol. 15, Casting — pattern and die draft

Route not stated, so the most permissive published minimum (0.5°) is applied — this under-reports rather than over-reports when the process is unknown.

[MAJOR] Abrupt section change (18 instances)

Worst: Section steps 8.54:1 at face 289 (24.4 mm against an adjacent 2.9 mm).

measured sectionRatio 2.364-8.54:1 threshold > 2:1

Faces: 22, 23, 47, 48, 71, 129, 131, 132, 146, 179, 202, 210, 217, 218, 256, 257, 259, 289

Fix: Blend the transition with a taper or a generous fillet so the section changes gradually; expect shrinkage porosity at the junction otherwise.

Source: ASM Handbook Vol. 15, Casting — design for solidification

A thick-to-thin ratio above ~2:1 across a junction concentrates solidification shrinkage and residual stress. Widely published as a 2:1 guideline; blended transitions are the standard remedy.

[MAJOR] Undercut face — needs a slide or a core (6 instances)

Worst: Face 186 sits at 135.0° to the draw (past 90°, so its normal opposes withdrawal) and cannot release on the main parting.

measured angleToDrawDeg 97.37-135° threshold > 90°

Faces: 61, 186, 241, 289, 295, 300

Fix: Re-orient the parting line, or price a slide/core for this feature and carry the die cost and cycle penalty explicitly.

Source: NADCA Product Specification Standards for Die Castings - Undercuts and moving die components

Any face whose normal opposes the draw cannot release on the main parting; it requires a slide, a loose piece or a core, each of which adds die cost and cycle time.

[MINOR] Sharp internal corner (14 instances)

Worst: Fillet at face 100 is R0.25 mm — effectively a sharp internal corner.

measured radiusMm 0.25-0.25mm threshold < 1mm

Faces: 89, 94, 96, 98, 100, 102, 106, 108, 110, 112, 114, 116, 118, 120

Fix: Increase the fillet toward half the adjoining wall thickness to relieve the stress concentration and the local hot spot.

Source: ASM Handbook Vol. 15, Casting — fillets and corner radii

Internal corners are stress raisers and hot spots. Common guidance is a fillet of roughly half the local wall; a radius under ~1 mm is treated as effectively sharp.

DFA - handling & insertion (Boothroyd method, geometric half)

Handling 1.63s + insertion 8.5s = 10.13s, 3.38x the 3s ideal part.

+0.5s Asymmetric envelope — the part must be oriented before insertion (bounding box 116×237×210 mm, no two dimensions within 5%)

+7s Fasteners approach from more than one direction — the part must be re-oriented or the fixture indexed during assembly (11 distinct hole axes across 44 holes)

Theoretical minimum part count, and therefore the Boothroyd design-efficiency index, requires the full assembly — this analysis covers one part, so no efficiency ratio is given.

Nesting and tangling risk is not assessed: it depends on how parts are presented in bulk, which the geometry alone does not determine.

Rotational symmetry is approximated from the bounding box, not from an inertia-tensor shape comparison.

What was NOT checked

No material family was confirmed, so any check whose threshold depends on the alloy or resin did not run.

Tolerance and datum callouts were not checked — they live on the drawing, not in the solid.

Parting-line position was assumed to lie perpendicular to the draw direction; a different parting choice changes which faces are undercuts.

Machining stock allowance on cast surfaces was not assessed — it is a process decision, not a measurable feature.

Gating and feeding layout was not evaluated; hot spots are reported, but where to place risers is a foundry decision.

Tolerance and surface-finish callouts were not checked — they are on the drawing, not in the solid, and they drive whether a feature is milled, ground or honed.

Tool access and fixturing were not simulated: a feature can be geometrically fine and still unreachable in the chosen setup.

Thread specifications were not verified against standard tap sizes.

Workholding and part distortion under clamping were not assessed.

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-gjl250	ID	
Grade / Specification	EN-GJL-250		UK iron foundry, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). Regional adj. ×0.880 (millSteel)
Region	China		
Net Finished Weight	2.6247 kg	kg	Weight in finished part
Gross Weight (stock)	4.0381 kg	kg	Net ÷ utilisation ratio
Scrap / Runner Weight	1.4133 kg	kg	Gross - Net
Material Utilisation	65.0%	%	Benchmark: casting 65–85 % (runners + risers)
Material Price	¥5.10	CNY /kg	UK iron foundry, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). Regional adj. ×0.880 (millSteel)
Scrap Recovery Price	¥0.96	CNY /kg	
Gross Material Cost	¥20.58	CNY	Gross × price/kg
Scrap Credit	-¥1.35	CNY	Scrap × recovery price
Consumables (core / wax / shell)	¥13.57	CNY	Per-part recurring
NET RAW MATERIAL COST	¥32.81	CNY	Gross - scrap credit + consumables
Data Confidence	Low		
Effective Date	2026-07		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Sand Casting — Moulding	Sand Casting Moulding Line	¥136.44	11.66	1	80.0%	¥33.14
Machining Setup (amortised)	HAAS VF-2 (3-axis Mill)	¥212.31	0.90	1	100.0 %	¥3.18
Milling — +Z (60 faces)	HAAS VF-2 (3-axis Mill)	¥212.31	1.10	1	85.0%	¥4.60
Milling — +X (51 faces)	HAAS VF-2	¥212.31	0.94	1	85.0%	¥3.90

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
(3-axis Mill)						
Milling — +Y (30 faces)	HAAS VF-2 (3-axis Mill)	¥212.31	0.55	1	85.0%	¥2.30
Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometry-measured]	CNC Drilling Centre	¥136.82	10.37	1	85.0%	¥27.83
TOTAL						¥74.95

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Sand Casting — Moulding	Skilled Machinist	¥71.49	1	11.66	92.0 %	¥15.10	¥48.25
Machining Setup (amortised)	Skilled Machinist	¥71.49	1	0.90	100.0%	¥1.07	¥4.26
Milling — +Z (60 faces)	Skilled Machinist	¥71.49	1	1.10	92.0 %	¥1.43	¥6.03
Milling — +X (51 faces)	Skilled Machinist	¥71.49	1	0.94	92.0 %	¥1.21	¥5.11
Milling — +Y (30 faces)	Skilled Machinist	¥71.49	1	0.55	92.0 %	¥0.71	¥3.01
Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometry-measured]	Skilled Machinist	¥71.49	1	10.37	92.0 %	¥13.44	¥41.27
TOTAL						¥32.97	¥107.92

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Boss	Ø25.0 × 17	2	Turning (external Ø)	1.28	—	No
Boss	Ø31.0 × 3	1	Turning (external Ø)	0.36	—	No
Boss	Ø31.0 × 26	1	Turning (external Ø)	0.82	—	No
Boss	Ø78.0 × 16	1	Turning (external Ø)	0.62	—	No
Hole / bore	Ø10.0 × 12 blind	1	Drilling (blind)	0.48	¥2.05	Yes
Hole / bore	Ø11.0 × 5 blind	1	Drilling (blind)	0.35	¥1.48	Yes
Hole / bore	Ø11.0 × 16 blind	1	Drilling (blind)	0.56	¥2.37	Yes

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Hole / bore	Ø12.0 × 2 blind	1	Drilling (blind)	0.29	¥1.23	Yes
Hole / bore	Ø12.5 × 16 blind	2	Drilling (blind)	1.14	¥4.83	Yes
Hole / bore	Ø12.5 × 25 blind	1	Drilling (blind)	0.75	¥3.18	Yes
Hole / bore	Ø15.0 × 19 blind	1	Drill + bore (blind)	1.26	¥5.32	Yes
Hole / bore	Ø15.0 × 25 blind	1	Drill + bore (blind)	1.50	¥6.36	Yes
Hole / bore	Ø63.0 × 3 blind	1	Drill + bore (blind)	0.45	¥1.92	Yes
Hole / bore	Ø68.0 × 4 blind	1	Drill + bore (blind)	0.47	¥2.00	Yes
Hole / bore	Ø68.0 × 37 blind	1	Drill + bore (blind)	2.42	¥10.28	Yes
Hole / bore	Ø71.0 × 3 blind	1	Drill + bore (blind)	0.48	¥2.05	Yes
Hole / bore	Ø75.0 × 6 blind	1	Drill + bore (blind)	0.50	¥2.12	Yes
Face	1544 mm²	1	Face milling (facing)	0.39	¥1.67	Yes
Face	1298 mm²	1	Face milling (facing)	0.36	¥1.53	Yes
Face	1184 mm²	1	Face milling (facing)	0.35	¥1.47	Yes
Face	1161 mm²	1	Face milling (facing)	0.34	¥1.46	Yes
Face	1012 mm²	1	Face milling (facing)	0.33	¥1.39	Yes
Face	1001 mm²	1	Face milling (facing)	0.33	¥1.38	Yes
Face	962 mm²	1	Face milling (facing)	0.32	¥1.36	Yes
Face	956 mm²	1	Face milling (facing)	0.32	¥1.36	Yes
Pocket	1544 mm² × 20	1	Pocket milling (rough + finish)	5.71	—	No
Pocket	1161 mm² × 34	1	Pocket milling (rough + finish)	7.26	—	No
Pocket	913 mm² × 34	1	Pocket milling (rough + finish)	5.82	—	No
Pocket	758 mm² × 7	1	Pocket milling (rough + finish)	1.18	—	No
Pocket	491 mm² × 68	1	Pocket milling (rough + finish)	6.37	—	No
TOTAL costed		22		13.40	¥56.81	Yes

10 detected feature(s) NOT costed

10 machinable feature instance(s) (29.4 min) were detected in the geometry but are NOT included in this cost — on a near-net part they may be as-cast. If they are machined, tick them in the Machined-Features panel and re-cost; this could add up to ¥124.69.

\$5 — Machine Rate Buildup

CNC Drilling Centre [mach-drill] · Confidence: Low · UK Tier-2 CNC drilling/tapping centre. Target £30/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥139366.52	¥43.55	4,000 hr/yr available
Maintenance	¥59728.51	¥18.67	
Energy	¥49577.02	¥15.49	
Floor Space	¥39819.00	¥12.44	
Indirect Support	¥79638.01	¥24.89	
Finance Cost	¥69683.26	¥21.78	
TOTAL — CNC Drilling Centre	¥437812.32	¥136.82	Effective hours: 3200/yr

HAAS VF-2 (3-axis Mill) [mach-haas-vf2] · Confidence: Low · HAAS VF-2 benchmark, UK Tier-2 machining shop | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @50% util	Notes
Depreciation	¥189140.27	¥94.57	4,000 hr/yr available
Maintenance	¥89592.76	¥44.80	
Energy	¥28919.93	¥14.46	
Floor Space	¥24886.88	¥12.44	
Indirect Support	¥47285.07	¥23.64	
Finance Cost	¥44796.38	¥22.40	
TOTAL — HAAS VF-2 (3-axis Mill)	¥424621.29	¥212.31	Effective hours: 2000/yr

Sand Casting Moulding Line [sand-cast-line] · Confidence: Low · UK foundry benchmark, Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @75% util	Notes
Depreciation	¥124434.39	¥47.40	3,500 hr/yr available
Maintenance	¥59728.51	¥22.75	
Energy	¥68856.97	¥26.23	
Floor Space	¥49773.76	¥18.96	
Indirect Support	¥39819.00	¥15.17	
Finance Cost	¥15554.30	¥5.93	
TOTAL — Sand Casting Moulding Line	¥358166.93	¥136.44	Effective hours: 2625/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf
material.pricePerKg	0.5632	£/kg	UK iron foundry, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). Regional adj. ×0.880 (millSteel)	mat-gjl250	Low
material.scrapRecoveryPricePerKg	0.1056	£/kg	UK iron foundry, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). Regional adj. ×0.880 (millSteel)	mat-gjl250	Low
rawMaterial.consumablesCostPerPart	1.5000	£	Per-part consumable (core/wax/shell)	mat-gjl250	Medium
Sand Casting — Moulding.machineRatePerHr	15.0771	£/hr	UK foundry benchmark, Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	sand-cast-line	Low
Sand Casting — Moulding.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Machining Setup (amortised).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Machining Setup (amortised).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Milling — +Z (60 faces).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Milling — +Z (60 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Milling — +X (51 faces).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Milling — +X (51 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Milling — +Y (30 faces).machineRatePerHr	23.4603	£/hr	HAAS VF-2 benchmark, UK Tier-2 machining shop Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-haas-vf2	Low
Milling — +Y (30 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometry-measured].machineRatePerHr	15.1182	£/hr	UK Tier-2 CNC drilling/tapping centre. Target £30/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-drill	Low

Field	Value	Unit	Source / Reference	Rate ID	Conf.
Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometry-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥198.06 ± 16.5%	¥166.70	¥196.38	¥232.04	moderate	12.8%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 14 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread; some detected features not costed (see §4C).

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.

Import duty and international freight (regional table is Ex-Works).

Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

Heat-treat (T6/T7), HIP and 100% NDT/X-ray are NOT in this cost — add them for structural / safety-critical parts.

Trim/degate, shot-blast/deflash and impregnation are only included where shown as operations.

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Sand Casting — Moulding: Cycle Time	0.19hr	-2.9%	+2.9%	¥11.67
Drilling — 14 holes (1×Ø10.0×12, 1×Ø11.0×5, 1×Ø11.0×16, 1×Ø12.0×2, 2×Ø12.5×16, 1×Ø12.5×25, 1×Ø15.0×19, 1×Ø15.0×25, 1×Ø63.0×3, 1×Ø68.0×4, 1×Ø68.0×37, 1×Ø71.0×3, 1×Ø75.0×6) [geometry-measured]: Cycle Time	0.17hr	-2.5%	+2.5%	¥9.98
Sand Casting — Moulding: Machine Rate	¥136.44/hr	-2%	+2%	¥8.02
Sand Casting — Moulding: Labour Rate	¥71.49/hr	-2%	+2%	¥7.98
Machining Setup (amortised): Labour Rate	¥71.49/hr	-2%	+2%	¥7.98
Milling — +Z (60 faces): Labour Rate	¥71.49/hr	-2%	+2%	¥7.98
Milling — +X (51 faces): Labour Rate	¥71.49/hr	-2%	+2%	¥7.98
Milling — +Y (30 faces): Labour Rate	¥71.49/hr	-2%	+2%	¥7.98

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥37.28	¥136.27	¥118.68	¥35.54	¥25.64	¥353.41	¥1.69	¥371.84	+88%
Germany (EUR)	¥38.40	¥143.08	¥191.81	¥37.32	¥35.34	¥445.94	¥1.94	¥468.54	+137%
France (EUR)	¥38.02	¥125.37	¥137.86	¥32.70	¥27.43	¥361.39	¥1.94	¥380.51	+92%
Spain (EUR)	¥37.28	¥119.92	¥86.91	¥31.28	¥20.47	¥295.85	¥1.94	¥312.22	+58%
Poland (PLN)	¥36.16	¥98.12	¥53.95	¥25.59	¥14.22	¥228.03	¥2.02	¥241.44	+22%
Turkey (TRY)	¥33.55	¥81.76	¥29.97	¥21.32	¥10.17	¥176.78	¥2.19	¥188.04	-5%
China (CNY)	¥32.81	¥74.95	¥32.97	¥19.55	¥19.23	¥179.50	¥2.44	¥198.06	Base
India (INR)	¥33.55	¥70.86	¥20.98	¥18.48	¥8.10	¥151.98	¥2.53	¥162.25	-18%
Mexico (MXN)	¥35.41	¥81.76	¥34.77	¥21.32	¥10.57	¥183.84	¥2.27	¥195.38	-1%
United States (USD)	¥37.28	¥115.83	¥155.84	¥30.21	¥25.21	¥364.37	¥2.11	¥383.58	+94%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

17.93 kgCO2e · 6.83 /kg · cradle-to-gate

Material	5.96 kgCO2e	Material factor	2.10 kg/kg (Steel)
Process	11.55 kgCO2e	Grid intensity	0.550 kg/kWh · 21.00 kWh
Logistics	0.42 kgCO2e	Total	17.93 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[WARNING] Machining / process cost above benchmark

Finding	Process cost at 37.8% of total is above the benchmark range of 8–18%. Cycle time or machine rate is elevated.
Impact	High · up to 10% potential saving
Benchmark	Process %: yours 37.8% vs industry 8–18%
Action 1	Reduce cycle time via feeds/speeds optimisation and toolpath strategy review
Action 2	Combine multiple operations into one setup to eliminate re-fixturing time

[WARNING] Elevated process cost — possible high reject/scrap rate

Finding	Process cost is 37.8% vs benchmark max of 18%. For castings, this may indicate a high reject rate (>5%) inflating effective machine time, poor OEE on the casting cell, or cycle time above industry benchmark for this alloy/weight range.
Impact	Medium · up to 8% potential saving
Action 1	Audit actual first-pass yield (FPY) data from the foundry — target >96% for HPDC Al
Action 2	Root-cause the dominant reject type: porosity, dimensional, surface, or cold shut

[WARNING] Low material utilisation — high scrap rate

Finding	Material utilisation at 65% is below the cast and machine benchmark of 82%. You are paying for 54% more raw material than ends up in the part.
Impact	Medium · up to 3% potential saving
Action 1	Redesign blank/preform geometry to minimise offcuts
Action 2	Optimise nesting/layout for sheet or profile stock

[WARNING] Low casting yield — excess runner/gating material

Finding	Casting yield of 65% means 54% of poured metal is returned as runner/gate scrap. HPDC benchmark is 65–75%; sand/gravity 75–85%. Scrap recovery credits are partial — you are paying for metal that doesn't end up in the part.
Impact	Medium · up to 2% potential saving
Benchmark	Casting Yield: yours 65.0% vs industry 65–85%
Action 1	Work with die designer to optimise runner/gate geometry and reduce pour weight
Action 2	Consider vacuum-assisted HPDC (vacural) to allow thinner gates and reduce gating volume

[INFO] Material cost is below benchmark range

Finding	Material at 16.6% is below the typical 35–60% range. Conversion cost is the dominant driver.
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Impact	Low
Benchmark	Material %: yours 16.6% vs industry 35–60%
Action 1	Focus optimisation efforts on process and labour efficiency

§12 — Cost-Reduction Opportunities (ranked by saving)

7 opportunities · combined ¥25.95/part (~13%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Material	Improve Material Utilisation via Near-Net-Shape	Current utilisation at 65%. Near-net-shape process or improved nesting targets 80–90%.	¥19.81	10.0 %	Medium Term	Medium
2	Material	Consumables Rationalisation (Cores, Patterns, Shell)	Per-part consumables are 41% of the material line. Core count, pattern life and shell recipe are all design-controllable.	¥11.88	6.0%	Medium Term	Medium
3	Material	Scrap Revenue Recovery at Index Prices	35% of the bought material leaves as chips, offcuts or skeleton. That scrap has an index price — the credit belongs in the piece price, not in the supplier's margin.	¥6.44	3.3%	Quick Win	Low
4	Design & Geometry	Part-Count / Feature Integration Review (Boothroyd)	6 operations suggest features or joined parts that may not need to exist separately. The Boothroyd minimum-part-count test asks of each: does it move, must it be a different material, must it be separate for assembly?	¥11.88	6.0%	Long Term	Medium
5	Design & Geometry	Tolerance and Surface-Finish Relaxation	Machining is 37.8% of the part. The tightest tolerance and finish callouts set the machine, the cycle and the inspection burden — reviewing them is the cheapest cost lever that exists.	¥7.92	4.0%	Quick Win	Low
6	Process & Automation	Unmanned / Lights-Out Running	Labour is 16.6% of cost on a machine-paced part with healthy OEE (87%) — the classic profile for bar-feed or pallet-pool unattended running.	¥9.89	5.0%	Medium Term	Medium
7	Process & Automation	Multi-Machine Manning	Operations are manned at up to 1.0 operators per machine while the machine paces the cycle. One operator tending two machines is the textbook correction.	¥8.24	4.2%	Quick Win	Low

§13 — Opportunity by Category

3 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Material	3	Improve Material Utilisation via Near-Net-Shape	¥19.81	¥38.13
Design & Geometry	2	Part-Count / Feature Integration Review (Boothroyd)	¥11.88	¥19.81
Process & Automation	2	Unmanned / Lights-Out Running	¥9.89	¥18.13

Item	What the costing assumed	How to close it
Unusually low material content — verify alloy grade	Material at 16.6% of part cost is below the 35% floor for castings. Verify alloy grade and net weight inputs.	Validate material weight and alloy price inputs against purchase orders

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Multi-Machine Manning
- Tolerance and Surface-Finish Relaxation
- Scrap Revenue Recovery at Index Prices

Long-Term Changes — Strategic Investment

- Part-Count / Feature Integration Review (Boothroyd)